Structures

Abstract Data Type

- You have seen many primitive data types like **`int, float, double, bool etc.`**

- An abstract data type (ADT) is a data type created by the programmer and is composed of one or more primitive data types.

Abstract Data Type

- So far you've written programs that keep data in individual variables.
- If you need to group items together, C++ allows you to create arrays.
- The limitation of arrays, however, is that all the elements must be of the same data type.
- Sometimes a relationship exists between different types of elements.

Abstract Data Type

Their definition statements do not make it clear that **they belong together**.

Variable Definition	**Data Held**
int empNumber;	Employee numb
string name;	Employee's name
double hours;	Hours worked
double payRate;	Hourly pay rate
double grossPay;	Gross pay

All these variables hold data about the same employee

Combining Data into Structures

- Structure: is like a container that allows multiple variables to be grouped together

- Variables can be of any type

```
struct structName
{
  dataType field1;
  dataType field2;
  . . .
};
```

Example `struct` Declaration

```
struct Student
{
   int studentID;
   string name;
   short int yearInSchool;
   double gpa;
};
```

structure name

structure members

- Organize related data (variables) into a nice neat package (single unit)

`struct` Declaration Notes

- Must have `;` after closing `}`
- `struct` names commonly begin with uppercase letter
- Multiple fields of same type can be in comma-separated list:

```
string name, address;
```

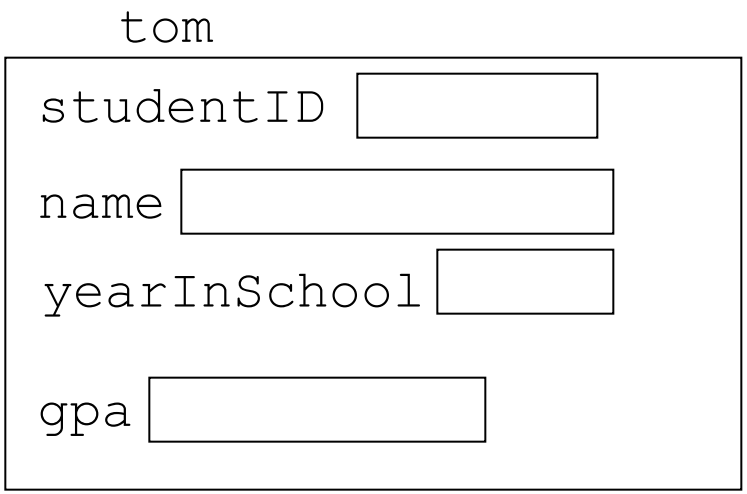
Creating `struct` Variables

- `struct` declaration does not allocate memory or create variables

- Must create a struct variable
- To create variables, use structure name as type name

```
Student tom;
```

structure name **variable name**

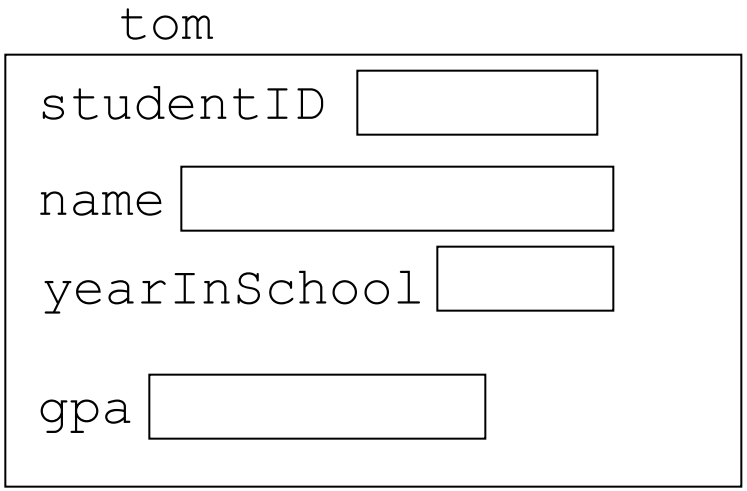


Creating `struct` Variables

- Must declare a structure before creating a structure variable

```
Student tom;
```

structure name **variable name**



Creating `struct` Variables – Another way

- Can also create a structure variable with its declaration

```
struct Student
{  int studentID;
   string name;
   short yearInSchool;
   double gpa;
} student1;
```

Creating `struct` Variables Two Ways

```
struct  Employee
{
    string      firstName;
    string      lastName;
    string      address;
    double      salary;
    int         deptID;
};

Employee  e1;
```

```
struct Student
{
    int studentID;
    string name;
    short yearInSchool;
    double gpa;
} s1, s2;
```

Accessing Structure Members

- Use the dot（.）operator to refer to members of **`struct`** variables:

```
cin >> s1.studentID;
s1.name = "Alex Stone";
s1.gpa = 3.75;
```

- Member variables can be used in any manner appropriate for their data type

Displaying a `struct` Variable

- To display the contents of a **`struct`** variable, must display each field separately, using the dot operator:

```
cout << s1; // won't work
cout << s1.studentID << endl;
cout << s1.name << endl;
cout << s1.yearInSchool;
cout << " " << s1.gpa;
```

Initializing a Structure

Values should be in the **same sequence** as the structure

```
struct Student
{  int studentID;
   string name;
   short yearInSchool;
   double gpa;
};
```

- `struct` variable can be initialized when created

```
Student s1 = {11465, "Joan", 2, 3.75};
```

Initializing a Structure

Can also be initialized member-by-member after definition:

```
s1.name = "Joan";
s1.gpa = 3.75;
```

More on Initializing a Structure

- May initialize only some members:

```
Student s1 = {14579};
```

- Cannot skip over members:

```
// illegal
Student s1 = {1234, "John", , 2.83};
```

More on Initializing a Structure

- You can also give default values inside a struct definition

```
struct Student
{
   int studentID = 0;
   string name = "";
   short yearInSchool = 1;
   double gpa = 1.0;
};
```

Accessing Structure Members

```
struct PayRoll  {
 int empNumber;
 string name;
 double hours;
 double payRate;
 double grossPay;
} emp1;
```

```
void main{

emp1.empNumber = 489;

emp1.name = "Jill Smith";

emp1.hours = 23;

emp1.payRate = 20;

emp1.grossPay = emp1.hours * emp1.payRate;
}
```

Comparing `struct` Variables

- Cannot compare `struct` variables directly:

```
if (s1 == s2) // won't work
```

- Instead, must compare on a field basis:

```
if (s1.studentID == s2.studentID)
```

Assigning `struct` Variables

- A structure variable can be assigned to another structure variable only if both are of same type

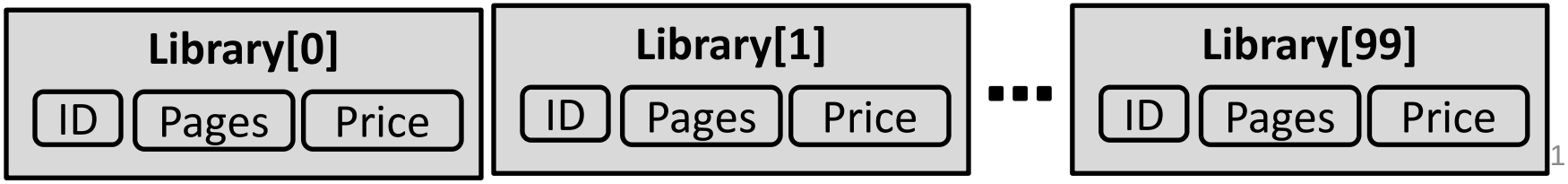
- A structure variable can be initialized by assigning another structure variable to it by using the assignment operator as follows:

```
Student s1 = { 1432, "Zoe", 3, 2.99} ;
Student s2 = s1;
```

Array of Structures

- An array of structures is a type of array in which each array element is a structure

```
struct Book
{       int ID;
        int Pages;
        float Price;
};
Book Library[100]; // declare array of structures
```



Arrays of Structures

```
const int NUM_STUDENTS = 20;
Student stuList[NUM_STUDENTS];
```

- Individual structures in an array accessible using subscript notation

- Fields within structures accessible using dot notation:

```
cout << stuList[5].studentID;
```

Array of Structures

```
struct  Book
{     int      ID;
      int      Pages;
      float  Price;
};
Book b[3]; //declare of array of structures
```

- Initializing can be at the time of declaration

  Book b[3] = { {1,275,70} , {2,600,90} , {3,786,100} };

- Or can be assigned values using cin:

```
cin >> b[0].ID;
cin >> b[0].Pages;
```

Structures as Function Arguments

- May pass members of `struct` variables to functions:

```
//function definition
float computeGPA(float gpa){
……————————
}

//function call
computeGPA(s1.gpa);
```

Structures as Function Arguments

- May pass entire `struct` variables to functions

  1. Pass-by-value
  2. Pass-by-reference
  3. Pass-using pointers

Structures as Function Arguments – Pass by Value

```
struct Rectangle {
    double length;
    double width;
    double area;
};
void changeRect(Rectangle r) {
    r.length = 5;
    r.width = 6;
    r.area = 30;
}
void main(){
Rectangle box = {1, 2, 2};
changeRect(box); }
```

A **copy of the struct box** is created and saved in the function parameter **r**

Structures as Function Arguments – Pass by Value

```
void changeRect(Rectangle r) {
    r.length = 5;
    r.width = 6;
    r.area = 30;
}
void main(){
Rectangle box = {1, 2, 2};
changeRect(box);

cout << box.length << endl; prints 1
cout << box.width << endl; prints 2
cout << box.area << endl; prints 2
}
```

Structures as Function Arguments – Pass by Reference

```
struct Rectangle {
    double length;
    double width;
    double area;
};
void changeRect(Rectangle &r) {
    r.length = 5;
    r.width = 6;
    r.area = 30;
}

Rectangle box = {1, 2, 2};
changeRect(box);
```

The **actual struct variable box is passed by reference**
(parameter **r** is just another name for box)

Structures as Function Arguments – Pass by Value

```
void changeRect(Rectangle &r) {
    r.length = 5;
    r.width = 6;
    r.area = 30;
}
void main(){
Rectangle box = {1, 2, 2};
changeRect(box);

cout << box.length << endl; prints 5
cout << box.width << endl; prints 6
cout << box.area << endl; prints 30
}
```

Structures as Function Arguments - Notes

- Passing a structure to a function by value can slow down a program, waste space

- Passing a structure to a function by reference will speed up program, but the function may change data in structure

- Using a `const` reference parameter allows read-only access to reference parameter, it is fast and does not waste space

Structures as Function Arguments – Pass by `const` Reference

```
void changeRect(const Rectangle &r) {
    r.length = 5;//ERROR! Cannot modify const
    r.width = 6; //ERROR! Cannot modify const
    r.area = 30; //ERROR! Cannot modify const
}
void main(){
Rectangle box = {1, 2, 2};
changeRect(box);


}
```

Structures as Function Arguments – Pass by `const` Reference

```
void showRect(const Rectangle &r) {
    cout << r.length << endl;
    cout << r.width << endl;
    cout << r.area << endl;
}
void main(){
Rectangle box = {1, 2, 2};
showRect(box);
}
```

Output:

```
1
2
2
```

Returning a Structure from a Function

- A Function can return a **`struct`**:

```
Student getStudentData();   // prototype
stu1 = getStudentData();    // call
```

- Function must define a local structure variable
  - for internal use
  - for use with **`return`** statement

Returning a Structure from a Function - Example

```
Student getStudentData()
{
    Student tempStu;

    cin >> tempStu.studentID;
    cin >> tempStu.yearInSchool;
    cin >> tempStu.gpa;

    return tempStu;
}
```

Practice Question 1

- Define a structure called **"car"**. The member elements of the car structure are:
  - » string Model;
  - » int Year;
  - » float Price

- Create an array of 30 cars. Get input for all 30 cars from the user. Then the program should display complete information (***Model***, ***Year***, ***Price***) of those cars only which are above 500,000 in price.

Practice Question 1

```
struct Car {
        string model;
        int year;
        float price;
};
void main() {
Car showroom[30]; //array of cars
for (int i = 0; i < 30; i++) {
        cin >> showroom[i].model;
        cin >> showroom[i].year;
        cin >> showroom[i].price;
}
for (int i = 0; i < 30; i++) {
        if (showroom[i].price > 500000) {
                cout << showroom[i].model<<" "<< showroom[i].year <<" "
                <<showroom[i].price;
                }
        }
}
```